

PATENTHUB-AI | OFFICIAL CLAIMS ENGINEERING DOCKET

Project: AI-Powered Adaptive Traffic Signal Optimization Using Real-Time Traffic Prediction | Domain: Artificial Intelligence | Exported: 2026-08-18

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1. PATENT CLAIMS SCHEDULE & TECHNICAL SPECIFICATION

Claim 1 (INDEPENDENT) [Independent] - Status: DRAFT

An automated intelligent ai-powered adaptive traffic signal optimization using real-time traffic prediction system, comprising: (a) a distributed sensor module configured to capture real-time operational environmental parameters; (b) an edge computing processor coupled to the sensor module to execute adaptive analytics; (c) a communication interface transmitting diagnostic status telemetries to a centralized registry.

Claim 2 (DEPENDENT) [Depends on Claim 1] - Status: DRAFT

The system of claim 1, wherein: the edge computing processor further executes real-time machine-learning anomaly detection based on statistical deviation thresholds.

Claim 3 (DEPENDENT) [Depends on Claim 1] - Status: DRAFT

The system of claim 1, wherein: the communication interface encrypts telemetries utilizing asymmetric cryptographic handshakes before dispatch.

2. PRELIMINARY FREEDOM-TO-OPERATE (FTO) MATRIX SUMMARY

No preliminary FTO claim charts generated yet.